

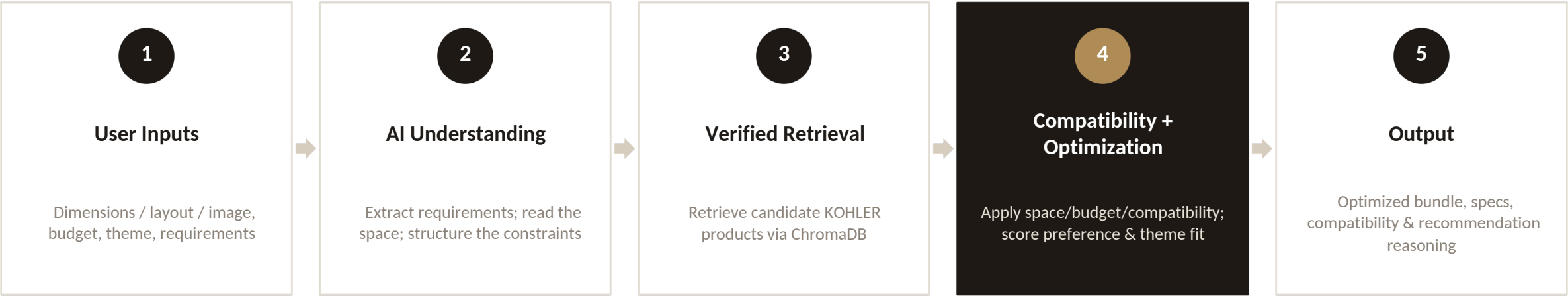
Core Approach

The Problem

Customers must juggle dimensions, layout, product compatibility, budget, function and style at once — product search alone doesn't solve this multi-constraint configuration problem.

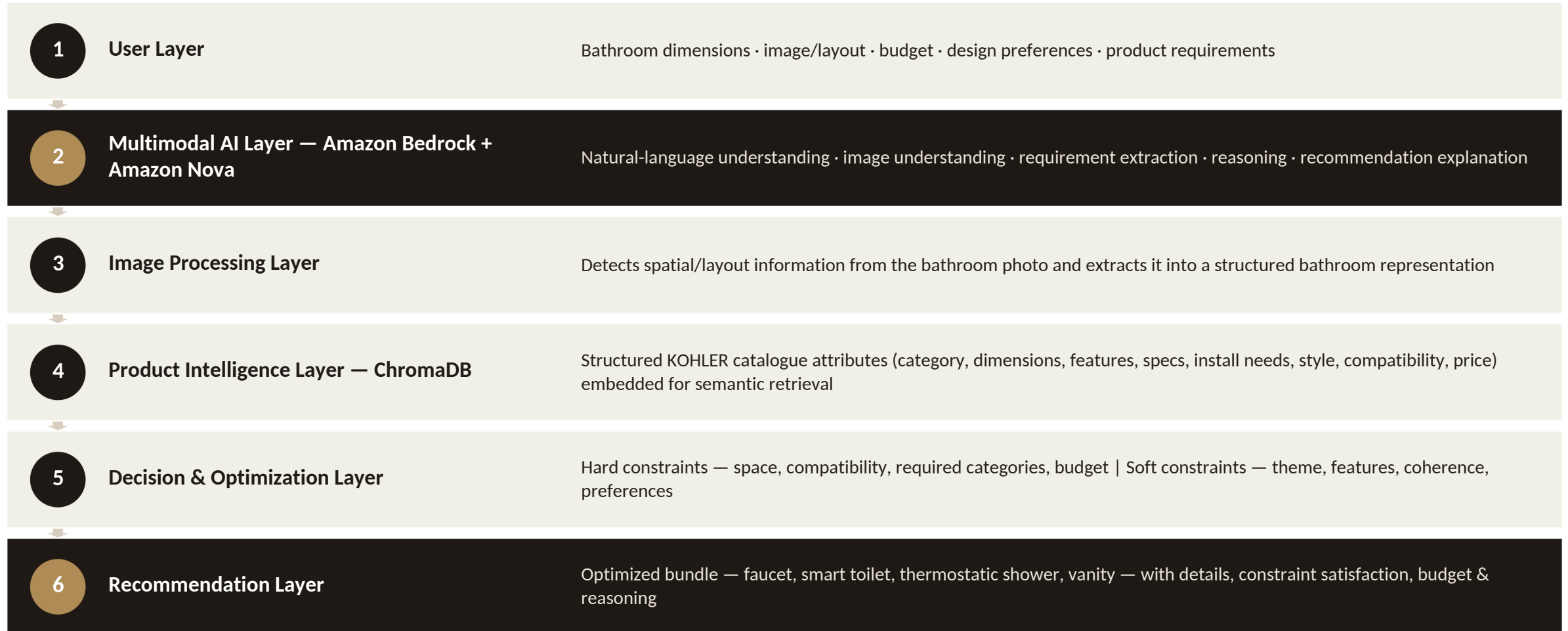
The Objective

An AI design assistant that turns customer requirements and bathroom-space information into a feasible, personalized, budget-aware KOHLER product bundle.



Central concept — AI understanding + verified catalogue retrieval + constraint optimization. Not free-form suggestion — bathroom planning treated as a constrained configuration problem.

System Architecture & End-to-End Flow



Architecture principle — generative AI handles understanding and reasoning; deterministic retrieval, compatibility checks and optimization handle structured product-selection decisions.

Technology Stack & Optimization Engine

Frontend

React 19 · Vite 7 · TypeScript · Tailwind CSS · shadcn/ui

Backend

Node.js · Express · tRPC · TypeScript · Zod · Drizzle ORM

Data

MySQL / TiDB · S3-compatible storage · KOHLER catalogue

AI & Intelligence

Amazon Bedrock/Nova (integration-ready) · Python/FastAPI · ChromaDB (integration-ready) · RAG

Core Engines

Catalogue retrieval · spatial intelligence · compatibility engine · colour intelligence · lookbook intelligence · constraint & budget optimization

DevOps & Testing

Docker · Vitest · Pytest · Git/GitHub

Decision Process

1. Candidate Retrieval
2. Hard-Constraint Filtering
3. Compatibility Filtering
4. Preference Scoring
5. Bundle Optimization
6. Final Recommendation

Optimization strategy — maximize overall bundle suitability (compatibility, space fit, preference & theme match, feature relevance) while penalizing budget deviation and constraint violations, subject to hard constraints.

Feasibility before preference — a visually preferred combination is never recommended if it violates space, budget or compatibility requirements.

Innovation Pitch & Differentiation

1

Multimodal Understanding
Reads bathroom images/layouts and natural-language requirements together

2

Verified Catalogue Grounding
Recommendations retrieved from structured KOHLER data—not model recall

3

Compatibility Reasoning
Checks spatial, technical and bundle-level fit for every candidate

4

Optimization
Searches feasible combinations rather than isolated products

Generic AI Recommendation



KOHLER AI Bathroom Designer



Example — 8 ft × 6 ft bathroom, ₹2,00,000 budget, Japanese Zen theme, needs a smart toilet, faucet, thermostatic shower and vanity → understands the brief, retrieves verified candidates, filters incompatible/space/budget-violating options, optimizes the combination, and outputs a complete bundle with reasoning.

Cost-Efficient & Scalable AI — Amazon Nova 2 Lite vs. Claude Fable 5.1 (Anthropic API benchmark)

\$0.30 vs \$10.00 /M in ~97% lower input cost

\$2.50 vs \$50.00 /M out ~95% lower output cost

Text + Image + Video native multimodal on Bedrock

From product search to intelligent bathroom configuration — understand the space, retrieve verified products, check compatibility, optimize the bundle, explain the decision.